UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/632,186	04/10/2024	Arik Tashie	06272-Tashie-NPV	5241

146458 7590 11/20/2025
NCCU School of Law-Patent Clinic
640 Nelson Street
Durham, NC 27707

EXAMINER	
LAWSON, STACY N	
ART UNIT	PAPER NUMBER
3678	
NOTIFICATION DATE	DELIVERY MODE
11/20/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

mafshar@nccu.edu
usptoincoming@nccu.edu

Office Action Summary	Application No. 18/632,186	Applicant(s) Tashie et al.	
	Examiner STACY N LAWSON	Art Unit 3678	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133).
  Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on April 10, 2024.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-17 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-17 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on April 10, 2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
Certified copies:
a)☐ All b)☐ Some** c)☐ None of the:
1.☐ Certified copies of the priority documents have been received.
2.☐ Certified copies of the priority documents have been received in Application No. _____.
3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date ______.

3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date ______.

4) ☐ Other: ______.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Objections

2. Claim 16 is objected to because of the following informalities: “a” should be added before “plurality” in line 3, and it appears that “iii.” should be inserted at the beginning of line 11. Appropriate correction is required.

3. Claim 17 is objected to because of the following informalities: “said” should be removed from line 3. Appropriate correction is required.

Claim Rejections - 35 USC § 112

4. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

5. Claims 1-17 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

6. Regarding claim 1, the wording of “A passive infiltration drainage system configured for transporting excess water from grassy areas in urban and suburban settings to desired storage areas comprising” in lines 1-2 is confusing. It is unclear which element is being referenced by the transitional phrase “comprising” because the phrase is positioned after “storage areas” but appears to be related to the “passive infiltration drainage system”. For purposes of examination, the examiner interprets “A

passive infiltration drainage system configured for transporting excess water from grassy areas in urban and suburban settings to desired storage areas comprising" to mean "A passive infiltration drainage system configured for transporting excess water from grassy areas in urban and suburban settings to desired storage areas, the passive infiltration drainage system comprising". Further, the wording of "a plurality of conveyance channels excavated in the landscape arranged in a network configuration" in lines 3-4 is confusing. There is insufficient antecedent basis for "the landscape". It is unclear which element is being referenced by the term "arranged" because the term is positioned after "landscape" but appears to be related to the "conveyance channels". For purposes of examination, the examiner interprets "a plurality of conveyance channels excavated in the landscape arranged in a network configuration" to mean "a plurality of conveyance channels excavated in a landscape and arranged in a network configuration". Further, the wording of "a plurality of infiltration tunnels excavated in the landscape that are connected to said plurality of conveyance channels configured for receiving and routing water flow" in lines 5-7 is confusing. The limitation suggests that the landscape is connected to the conveyance channels. It is unclear which element is being referenced by the phrase "configured for" because the phrase is positioned after "conveyance channels" but appears to be related to the "infiltration tunnels". For purposes of examination, the examiner interprets "a plurality of infiltration tunnels excavated in the landscape that are connected to said plurality of conveyance channels configured for receiving and routing water flow" to mean "a plurality of infiltration tunnels excavated in the landscape and connected to said plurality of conveyance channels, wherein the plurality of infiltration tunnels are configured for receiving and routing water flow". Claims 2-15 are rejected for depending from a rejected claim.

7. Regarding claim 6, the limitation "may be" in line 2 renders the claim indefinite because it is not clear if the recitation that follows is required or is optional. For purposes of examination, the examiner interprets "may be" to mean "are capable of being".

8. Regarding claim 10, the wording of “said sand top layer is about 0.5 mm to 2 mm” in lines 1-2 is confusing. Does this refer to a thickness? A length or width? A sediment size? For purposes of examination, the examiner interprets “said sand top layer is about 0.5 mm to 2 mm” to mean “said sand top layer comprises sand with a sediment size of about 0.5 mm to 2 mm”.

9. Claim 13 recites the limitation “said bentonite clay” in line 1. There is insufficient antecedent basis for this limitation in the claim.

10. Claim 14 recites the limitation “said conductive hydraulic substrate” in lines 1-2. There is insufficient antecedent basis for this limitation in the claim.

11. Regarding claim 15, it is unclear whether “an optional water discharge outlet” in lines 1-2 is the same as or different than, and in addition to, “an optional water discharge outlet” in claim 1 because of the double positive recitation of “an optional water discharge outlet”. For purposes of examination, the examiner interprets “an optional water discharge outlet” to mean “the optional water discharge outlet”.

12. Regarding claim 16, the wording of “an optional water discharge outlet” in line 5 is confusing. The claim positively recites that the conveyance channels are configured to empty said water into said optional water discharge outlet. How can the conveyance channels perform this step without the existence of the water discharge outlet? For purposes of examination, the examiner interprets“ an optional water discharge outlet” to mean “a water discharge outlet”. Further, it is unclear whether “a bedrock or saprolite” in lines 11-12 is the same as or different than, and in addition to, “bedrock or saprolite” in line 4 because of the double positive recitation of “bedrock or saprolite”. For purposes of examination, the examiner interprets “a bedrock or saprolite” to mean “the bedrock or saprolite”.

13. Regarding claim 17, the wording of “A method of draining water from a land surface to a desired storage area comprising the steps of” in lines 1-2 is confusing. It is unclear which element is being referenced by the transitional phrase “comprising” because the phrase is positioned after “storage area” but appears to be related to the “method”. For purposes of examination, the examiner interprets “A

method of draining water from a land surface to a desired storage area comprising the steps of" to mean "A method of draining water from a land surface to a desired storage area, the method comprising the steps of". Further, it is unclear whether "conveyance channels" in line 5 are the same as or different than, and in addition to, "a plurality of conveyance channels" in lines 3-4 because of the double positive recitation of "conveyance channels". For purposes of examination, the examiner interprets "conveyance channels" to mean "the conveyance channels". Finally, the wording of "emptying said water from conveyance channels into an optional water discharge outlet" in lines 5-6 is confusing. The claim positively recites the step of emptying the water into an optional water discharge outlet therefore how can the water discharge outlet be optional? If the water discharge outlet were not present then the step could not be completed. For purposes of examination, the examiner interprets "emptying said water from conveyance channels into an optional water discharge outlet" to mean "emptying said water from conveyance channels into a water discharge outlet".

14. Claim 16 is rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being incomplete for omitting essential structural cooperative relationships of elements, such omission amounting to a gap between the necessary structural connections. See MPEP § 2172.01. The omitted structural cooperative relationships are: the relationship between (a) the conveyance channels and water discharge outlet and (b) the infiltration tunnels and bedrock/saprolite. There is no relationship recited between the channels/outlet and the tunnels/bedrock therefore these elements could be part of entirely different systems. These relationships are essential for the drainage system to properly function when performing its intended use of draining water from land.

Claim Rejections - 35 USC § 103

15. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention

> and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

16. Claims 1-17 (as best understood) are rejected under 35 U.S.C. 103 as being unpatentable over des Garennes et al (US 2017/0167098) alone.

17. Regarding claim 1, des Garennes discloses a passive infiltration drainage system configured for transporting excess water from grassy areas in urban and suburban settings to desired storage areas (e.g. 100, Fig. 1, paragraphs 0002-0003), the passive infiltration drainage system comprising: a. a plurality of conveyance channels excavated in the landscape arranged in a network configuration (e.g. 110/120, Fig. 1, paragraphs 0003 and 0025, wherein the portions of 110/120 above the instances of 130/140/150 form a network of channels); b. a plurality of infiltration tunnels excavated in the landscape that are connected to said plurality of conveyance channels configured for receiving and routing water flow (e.g. 130, Fig. 1, paragraph 0025); and c. an optional water discharge outlet for collecting water from said plurality of conveyance channels (e.g. similar to 330, Fig. 3, paragraph 0032). des Garennes further discloses an element for removing the water from the soil profile (e.g. 150, Fig. 1, paragraph 0021) but does not disclose that the plurality of infiltration tunnels are connected to bedrock or saprolite. The examiner takes official notice that bedrock and saprolite are notoriously well known natural materials in the art. It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to extend the element of des Garennes down to bedrock or saprolite (thereby connecting the infiltration tunnels to the bedrock or saprolite via the element) for the expected benefit of ensuring that the water is carried deep enough to effectively drain the upper layers of soil as desired.

18. Regarding claim 2, des Garennes further discloses that said plurality of conveyance channels comprises sand (e.g. paragraph 0022).

19. Regarding claim 3, des Garennes further discloses that said plurality of conveyance channels are spaced about three to five feet apart with a width of about one foot (e.g. paragraph 0025 wherein the channels are positioned above the drains 130 which are spaced three feet apart and Fig. 1 wherein a width of one foot of 110/120 is considered the channel) but des Garennes does not disclose that said plurality of conveyance channels are a depth of about one to two feet. It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to contrive any number of desirable ranges for the depth limitation disclosed by Applicant, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or workable ranges involves only routine skill in the art. Further, it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. Finally, Applicant has not disclosed that this depth provides an advantage, is used for a particular purpose, or solves a stated problem.

20. Regarding claim 4, des Garennes does not disclose that said plurality of conveyance channels are dug to a depth of about one to two feet. It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to contrive any number of desirable ranges for the depth limitation disclosed by Applicant, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or workable ranges involves only routine skill in the art. Further, it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. Finally, Applicant has not disclosed that this depth provides an advantage, is used for a particular purpose, or solves a stated problem.

21. Regarding claim 5, des Garennes further discloses that said plurality of conveyance channels are dug to a width of one foot (e.g. Fig. 1 wherein a width of one foot of 110/120 is considered the channel).

22. Regarding claim 6, des Garennes further discloses that said plurality of infiltration tunnels are drilled gradually upslope when adjacent to one or more structures that may be negatively impacted by excess water (e.g. similar to Fig. 3 wherein the tunnels are formed upslope).

23. Regarding claim 7, des Garennes does not disclose that said plurality of infiltration tunnels are drilled to a radius of 3 inches, and to a depth of 6 feet. It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to contrive any number of desirable ranges for the radius and depth limitations disclosed by Applicant, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or workable ranges involves only routine skill in the art. Further, it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. Finally, Applicant has not disclosed that this radius and depth provide an advantage, are used for a particular purpose, or solve a stated problem.

24. Regarding claim 8, des Garennes further discloses that said plurality of infiltration tunnels are drilled directly under impervious surfaces (e.g. paragraph 0024, or paragraph 0023 wherein clay soils are part of 120 and are impervious inasmuch as the clay surface of the instant application is impervious).

25. Regarding claim 9, des Garennes further discloses a sediment filter positioned in each of said plurality of conveyance channels, wherein said sediment filter comprises a sand top layer (e.g. 110, Fig. 1, paragraph 0022) and clay base (e.g. 120, paragraph 0023 wherein clay is part of 120).

26. Regarding claim 10, des Garennes further discloses optimizing particle size of said sand top layer (e.g. paragraph 0022) but does not disclose that said sand top layer is about 0.5 mm to 2 mm. It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to contrive any number of desirable ranges for the particle size limitation disclosed by Applicant, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or workable ranges involves only routine skill in the art. Further, it

has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. Finally, Applicant has not disclosed that this particle size provides an advantage, is used for a particular purpose, or solves a stated problem.

27. Regarding claim 11, des Garennes further discloses that said sand top layer comprises coarse sand (e.g. paragraph 0003) and discloses optimizing said sand top layer (e.g. paragraph 0022) but does not disclose that said sand top layer comprises coarse sand and rounded sand. It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to use a combination of coarse and rounded sand for the sand top layer of des Garennes because a change in the shape of a prior art device is a design consideration within the skill of the art and the combination of coarse sand and rounded sand would provide the expected benefit of taking advantage of the properties of multiple shapes of sand.

28. Regarding claim 12, des Garennes does not disclose that said clay is bentonite clay with a sediment size of 0.8 µm to 2000 µm. The examiner takes official notice that bentonite clay is notoriously well known in the art. It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to have bentonite clay because such is a known clay typically found in earth formations. It further would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to contrive any number of desirable ranges for the sediment size limitation disclosed by Applicant, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or workable ranges involves only routine skill in the art. Further, it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. Finally, Applicant has not disclosed that this sediment size provides an advantage, is used for a particular purpose, or solves a stated problem.

29. Regarding claim 13, des Garennes does not disclose that said bentonite clay is compacted to a thickness of 1 inch. It would have been obvious to a person having ordinary skill in the art, before the

effective filing date of the claimed invention, to contrive any number of desirable ranges for the thickness limitation disclosed by Applicant, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or workable ranges involves only routine skill in the art. Further, it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. Finally, Applicant has not disclosed that this thickness provides an advantage, is used for a particular purpose, or solves a stated problem.

30. Regarding claim 14, des Garennes further discloses that said conductive hydraulic substrate is saprolite (e.g. as explained above in claim 1).

31. Regarding claim 15, des Garennes further discloses an optional water discharge outlet directing collected water out of the drainage system (e.g. similar to 330, Fig. 3, paragraph 0032) but does not explicitly disclose that the water is directed into a body of water. It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to direct the water into a body of water for the expected benefit of using a natural water source.

32. Regarding claim 16, des Garennes discloses a passive infiltration drainage system (e.g. 100, Fig. 1, paragraphs 0002-0003) comprising: a. a plurality of conveyance channels (e.g. 110/120, Fig. 1, paragraph 0025, wherein the portions of 110/120 above the instances of 130/140/150 form a plurality of channels); b. plurality of infiltration tunnels (e.g. 130, Fig. 1, paragraph 0025); and d. an optional water discharge outlet (e.g. similar to 330, Fig. 3, paragraph 0032); wherein: i. said system is configured to drain water from a land surface to said plurality of conveyance channels (e.g. Fig. 1, paragraphs 0021 and 0025); ii. said plurality of conveyance channels are configured to empty said water into said optional water discharge outlet (e.g. Fig. 3, similar to paragraph 0032); and said plurality of infiltration tunnels are configured to drain said water (e.g. Fig. 1, paragraph 0025). des Garennes does not disclose that the water is drained to a bedrock or saprolite. The examiner takes official notice that bedrock and saprolite are notoriously well known natural materials in the art. It would have been obvious to a person having

ordinary skill in the art, before the effective filing date of the claimed invention, to drain the water of des Garennes down to bedrock or saprolite for the expected benefit of ensuring that the water is carried deep enough to effectively drain the upper layers of soil as desired.

33. Regarding claim 17, des Garennes discloses a method of draining water from a land surface to a desired storage area (e.g. paragraph 0002) comprising the steps of: i. draining said water from a land surface to a plurality of conveyance channels (e.g. 110/120, Fig. 1, paragraphs 0021 and 0025, wherein the portions of 110/120 above the instances of 130/140/150 form a plurality of channels); and ii. emptying said water from conveyance channels into an optional water discharge outlet (e.g. 150, Fig. 1, or similar to 330, Fig. 3, paragraph 0032).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STACY N LAWSON whose telephone number is (571)270-7515. The examiner can normally be reached Mon-Fri 9am-3pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner’s supervisor, Amber Anderson can be reached at 571-270-5281. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/S.N.L./
Examiner, Art Unit 3678

/AMBER R ANDERSON/
Supervisory Patent Examiner, Art Unit 3678